

PIVOTAL TUNING EDITING: TOWARDS WRINKLE EDITING WITH GANS

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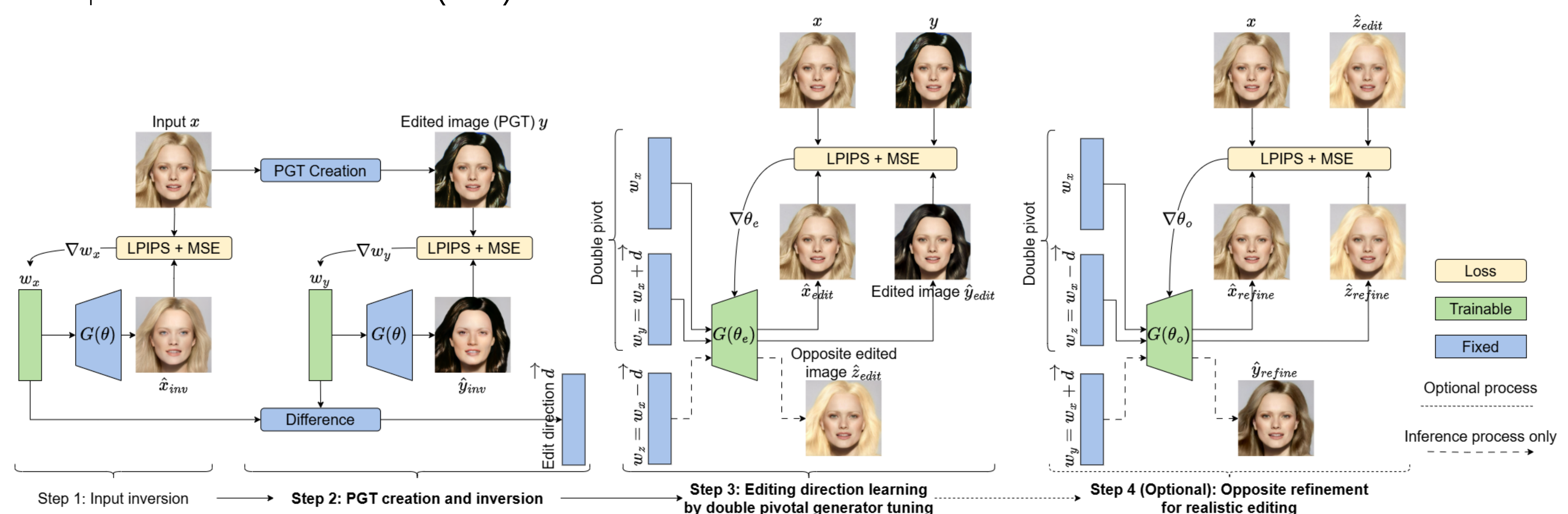
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ABSTRACT

GANs enable **image editing** by manipulating image features. However, these manipulations still lack **disentanglement**. For example, when a specific wrinkle is edited, other age-related features or facial expressions are often changed as well. The presented approach is based on **two pivot images** that allow learning an editing direction for an input image. These pivots are based on a real image (the input) and a synthetic modification of the real image along the desired editing direction. Although our primary focus is on **wrinkle editing**, our method can be extended to other editing tasks, such as **hair color** or **lipstick editing**.

ARCHITECTURE

The main idea (step 3) is to **fine-tune** the GAN generator $G(\theta)$ using two fixed latent variables. These two latent variables, called **double pivots**, define the desired editing direction. The first pivot (step 1) w_x is obtained by inverting the input x . The second pivot w_y is obtained by inverting a modified version of the input along the desired direction (step 2). We call this modified input **Pseudo-Ground Truth (PGT)**.



QUALITATIVE RESULTS



QUANTITATIVE RESULTS

		Vertical Container			Vertical Container	
		IC ↓	LD _o ↓	MSE _o ↓	SSIM ↑	FID ↓
Lion's Wrinkle	Resefa	0.32	154	0.89	0.859	89
	StyleMapGAN	0.27	29	1.43	0.963	119
	SOAT	0.08	51	0.09	0.995	76
	PTE (Ours)	0.11	67	0.21	0.969	31
Lipstick Editing	Resefa	0.48	129	0.94	0.842	66
	StyleMapGAN	0.45	66	2.00	0.940	95
	StyleClip	0.16	68	0.29	0.975	69
	FEAT	0.23	103	1.37	0.724	77
	PTE (Ours)	0.15	65	0.34	0.965	28